The Future of Healthcare

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Preventive Healthcare

AIMS Lab
Advanced Intelligent Multidisciplinary Systems Lab

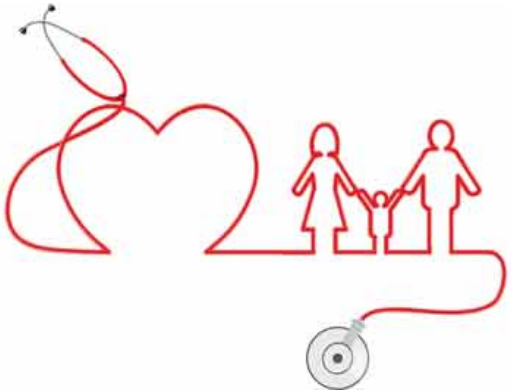
UIU
UNITED INTERNATIONAL UNIVERSITY
QUEST FOR EXCELLENCE

CEMS Global

Healthcare

- **Healthcare**
  - *Health care is the maintenance or improvement of health via the **prevention, diagnosis, and treatment** of disease, illness, injury, and other physical and mental impairments in human beings.*
  - *All goods and services designed to promote health, including **preventive, curative and palliative** interventions, whether directed to individuals or to populations*
  - *The act of taking **preventative or necessary medical procedures** to improve a person's well-being.*

Healthcare is provided through healthcare systems



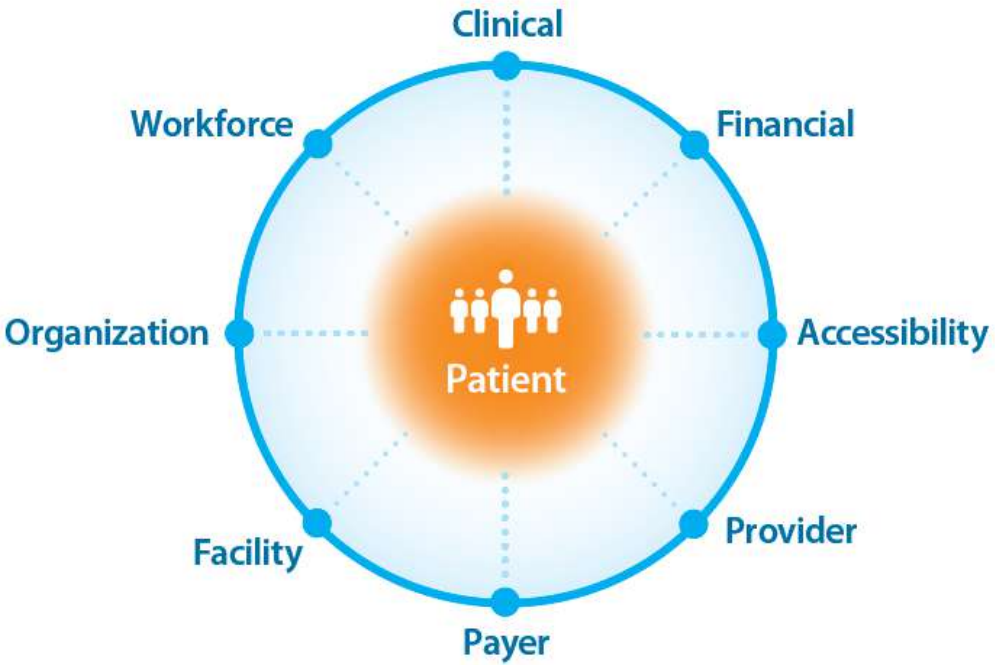
Healthcare Systems

- **Healthcare Systems**
  - All activities whose primary purpose is to promote, restore or maintain health
  - It includes work done in providing primary care, secondary care, and tertiary care, as well as in **public health**
  - Services are typically offered through a health care system made up of hospitals and physicians



Components of Healthcare Systems





Global: Healthcare Crisis

Mature Economies: Rising healthcare costs is an ongoing challenge

Challenges in Developed Economies



Rising healthcare costs



Ageing population



Increasing chronic disease prevalence

Developing Economies: Lack of healthcare access is a growing problem

Challenges in Developing Economies



Lack of healthcare access



Increasing chronic disease prevalence



Lack of infrastructure

Global: Healthcare Crisis

Non-communicable diseases account for **67% of deaths** in low middle income countries but only **1% of health funding** addresses them



Source: Institute for Health Metrics and Evaluation

% of deaths in low- and middle-income countries in 2015

% of total health funding in 2015 including government, philanthropy, and international organizations

Healthcare Scenario

Key contributors to increasing NCDs:

Rapid urbanization



Unhealthy lifestyles



Healthcare Crisis in Bangladesh

No Referral System

Healthcare Service into Unknown

Not Affordable

67% out of Pocket Expense

No Health Record

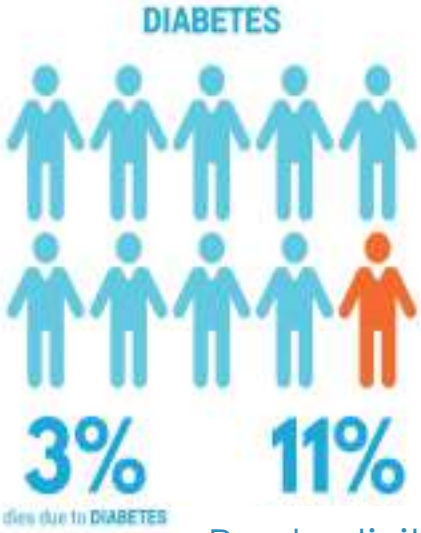
Lack Proper Treatment

61% Deaths

Non-communicable diseases

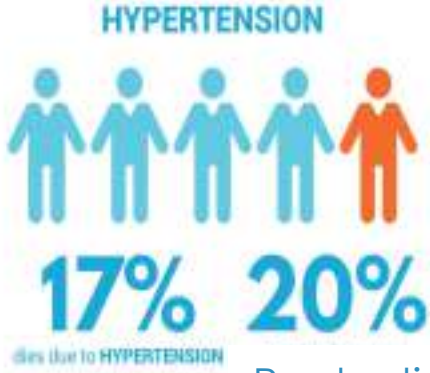
PROBLEM

Non-Communicable Diseases (NCDs)



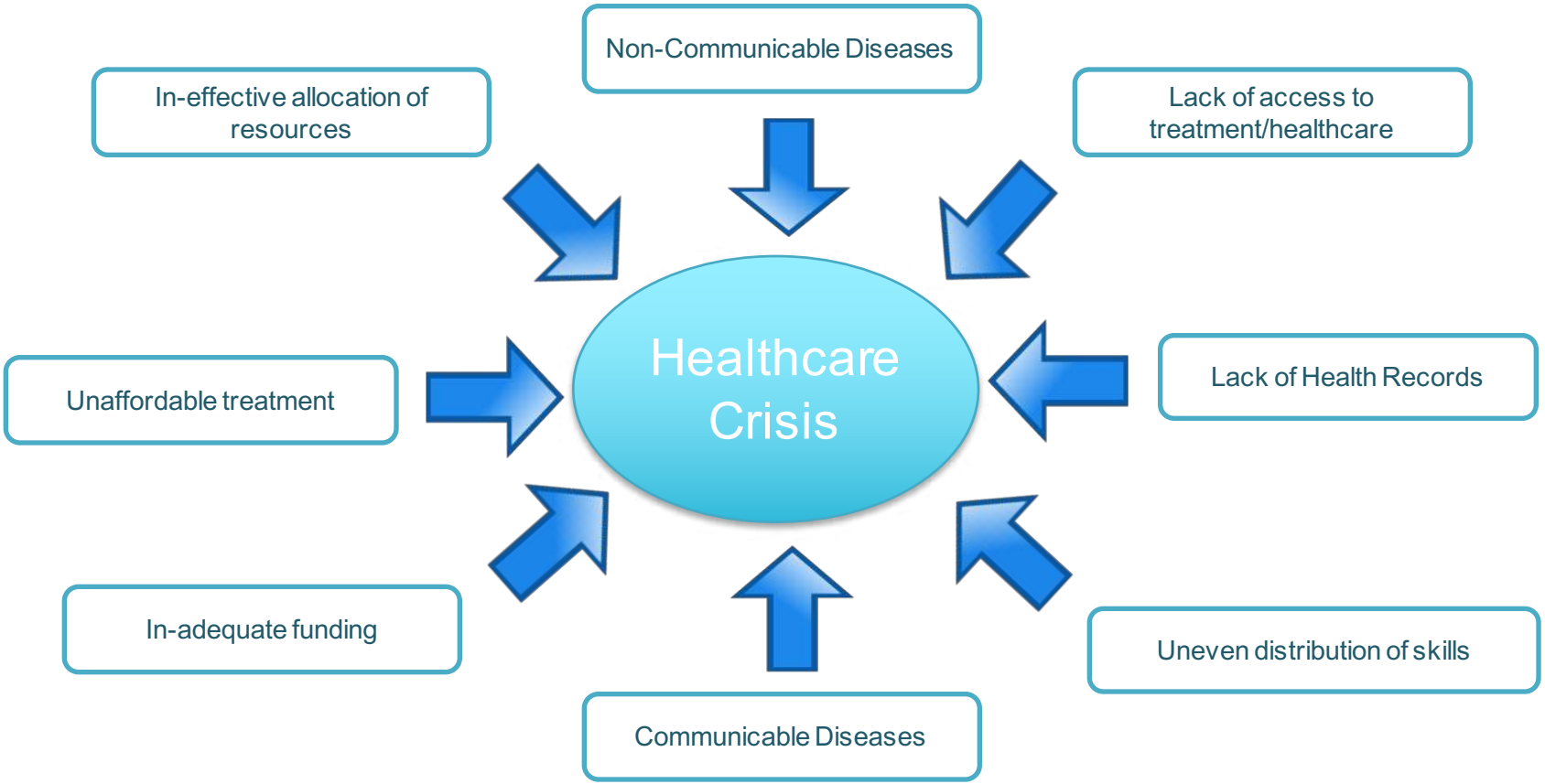
Productivity **Loss**





Productivity **Loss**

Healthcare Crisis



Prevention is Better Than Cure



Preventive Care

9/10

strokes can be prevented

80%
Heart Attack

73%
Diabetes

5x

knowing your vitals
will reduce the health
risk & cost by **FIVE** times

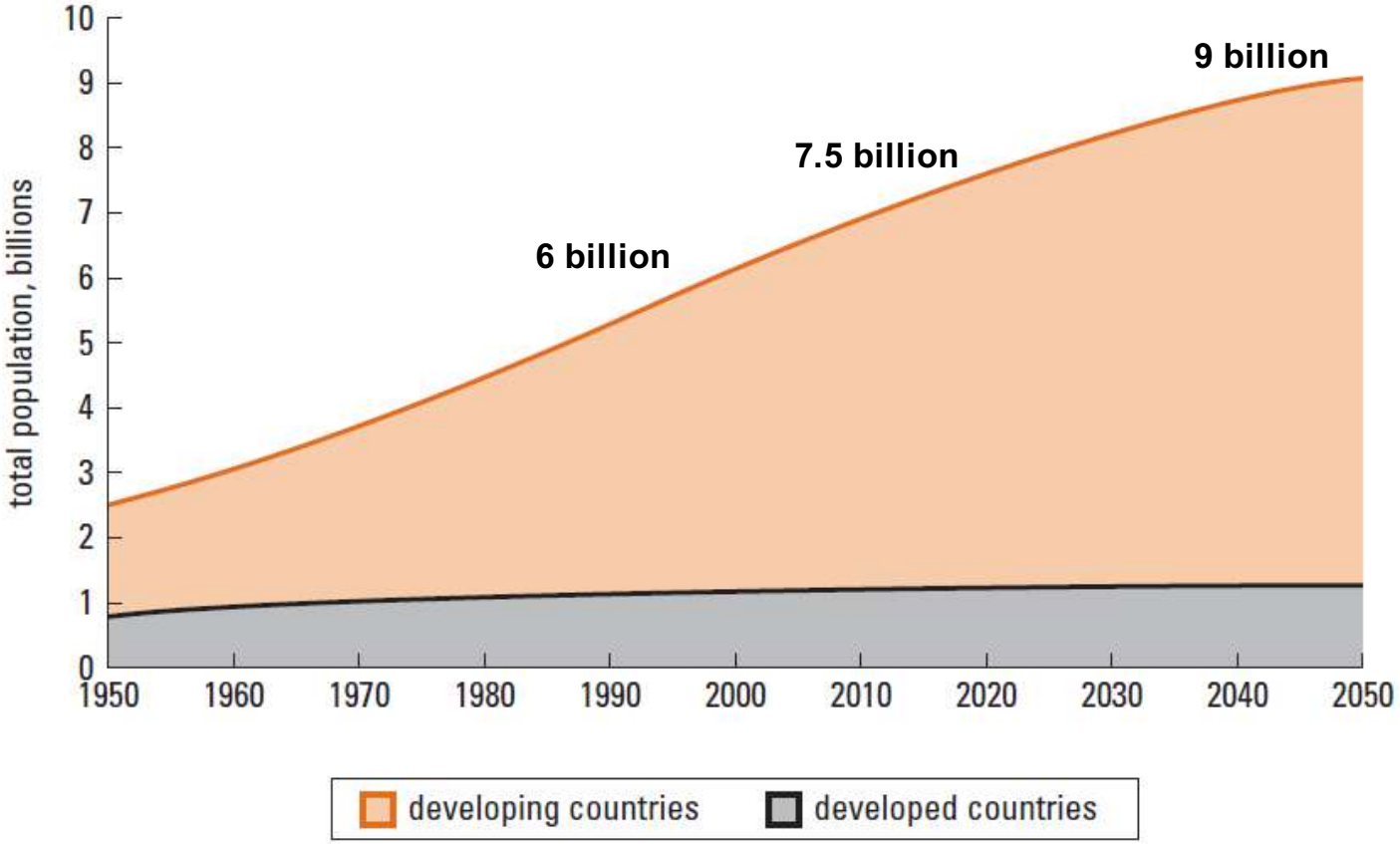
Universal Health Coverage



The United Nations **Sustainable Development Goals** that all UN Member States have agreed to try to **achieve Universal Health Coverage by 2030**. This includes financial risk protection, access to quality essential health-care services and access to safe, effective, quality and affordable essential medicines and vaccines for all.

Healthcare Finance

Health financing, demographic changes and disease burdens



Source: United Nations

Healthcare Finance





Patients

Payers

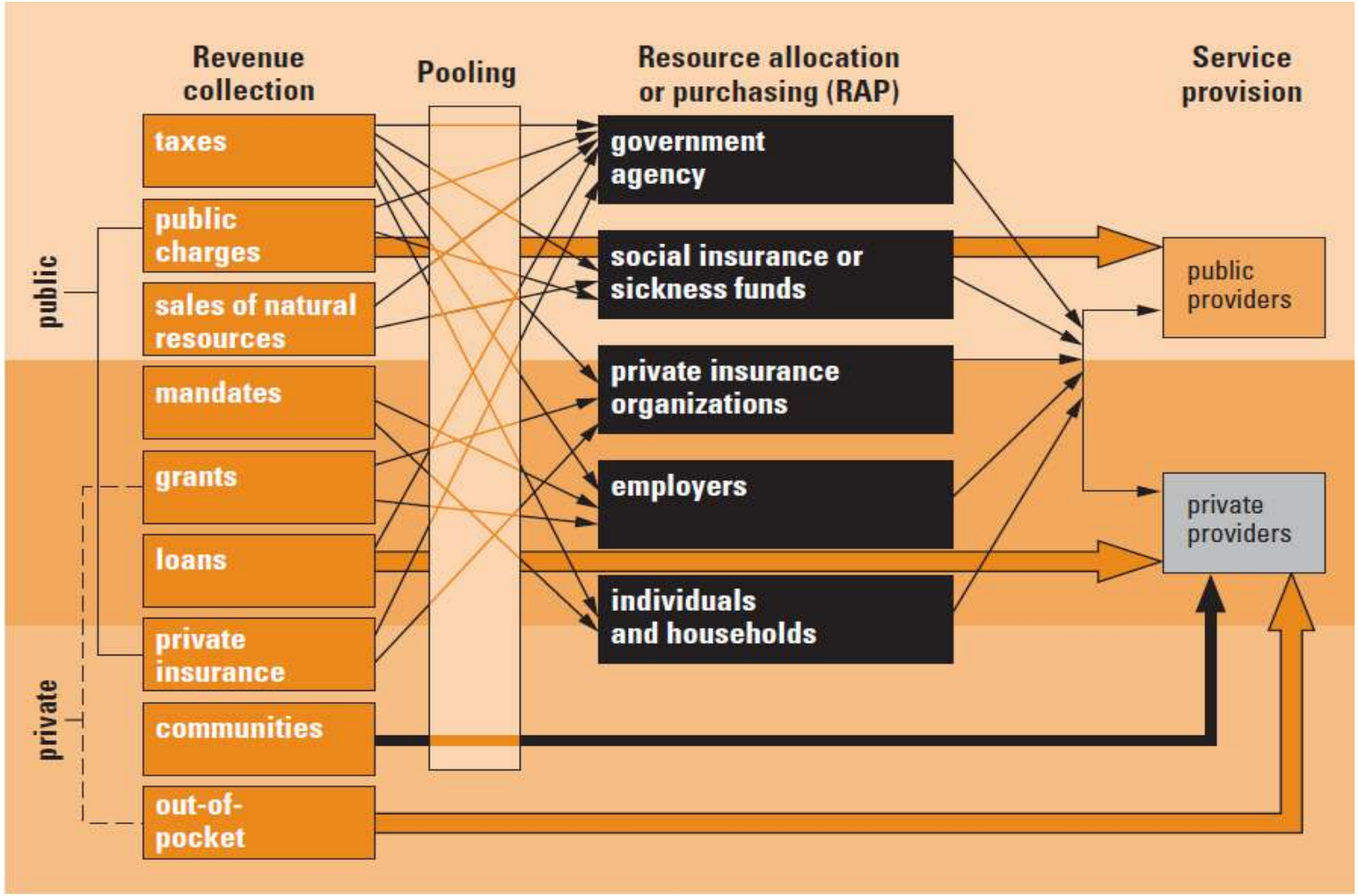
Providers

Patients periodically pay premiums (or taxes) to the payer.

The payer pays for a portion of the patient's treatment.

When a patient goes to the doctor, they pay a co-pay.

Healthcare Finance



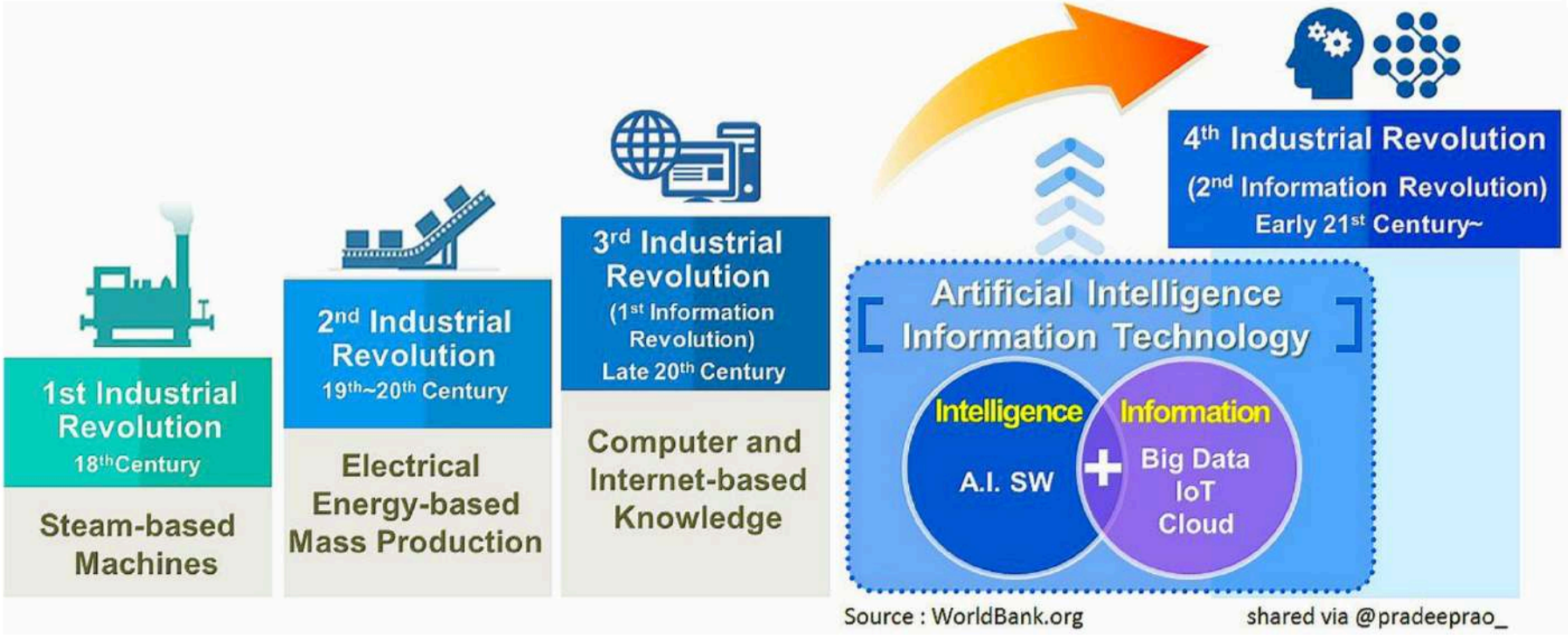
Digital Technology for Healthcare







Utilization of 4th Industrial Revolution in Healthcare



DRIVERS OF HEALTHCARE TECHNOLOGY



REGULATORY REQUIREMENTS

- Payer reform driven by quality vs. quantity
- Transparency of healthcare costs
- Provider collaboration
- Preventative medicine



NEW DELIVERY MODELS

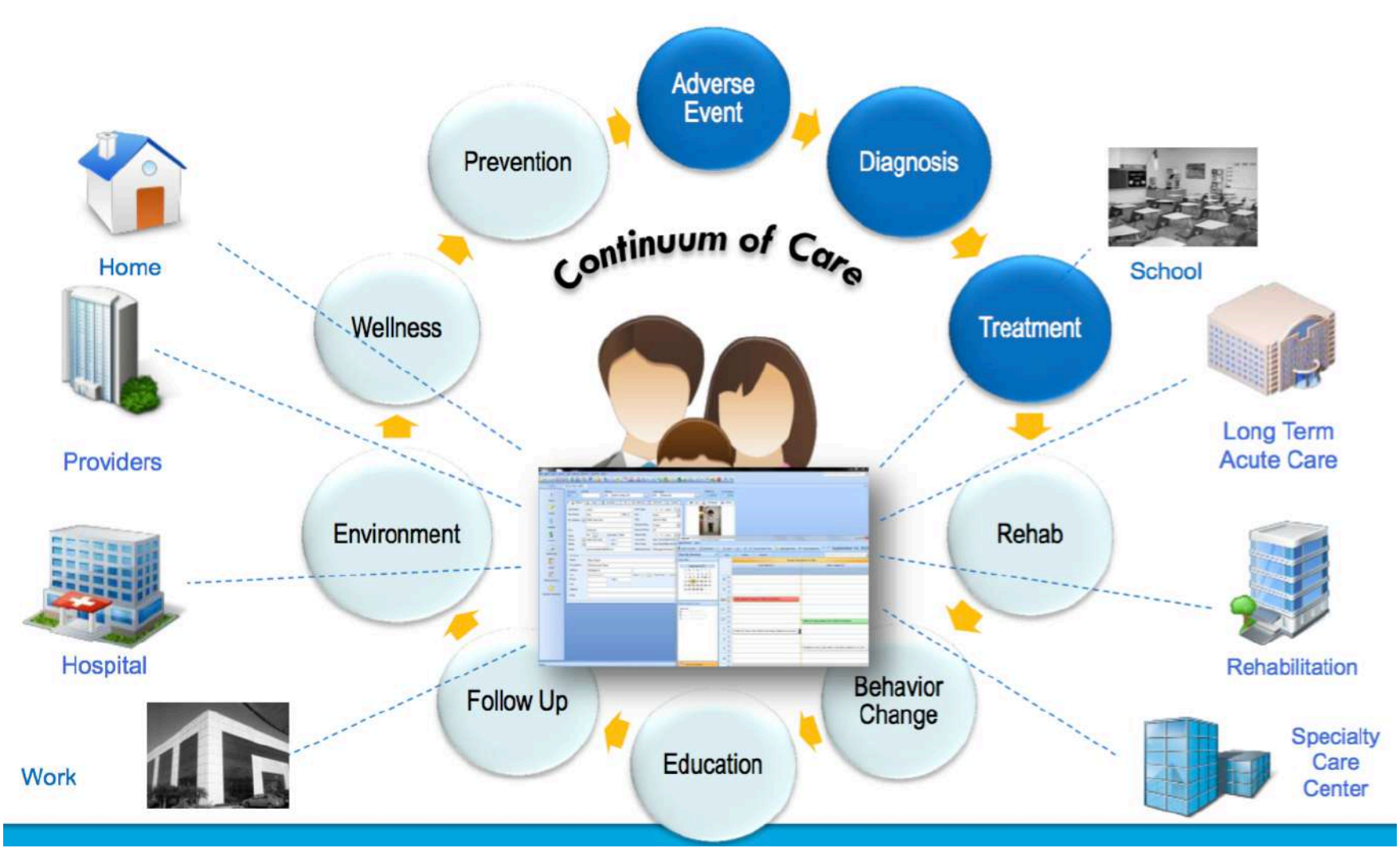
- Retail health clinics
- Shift from hospital to ambulatory services



PATIENT ENGAGEMENT

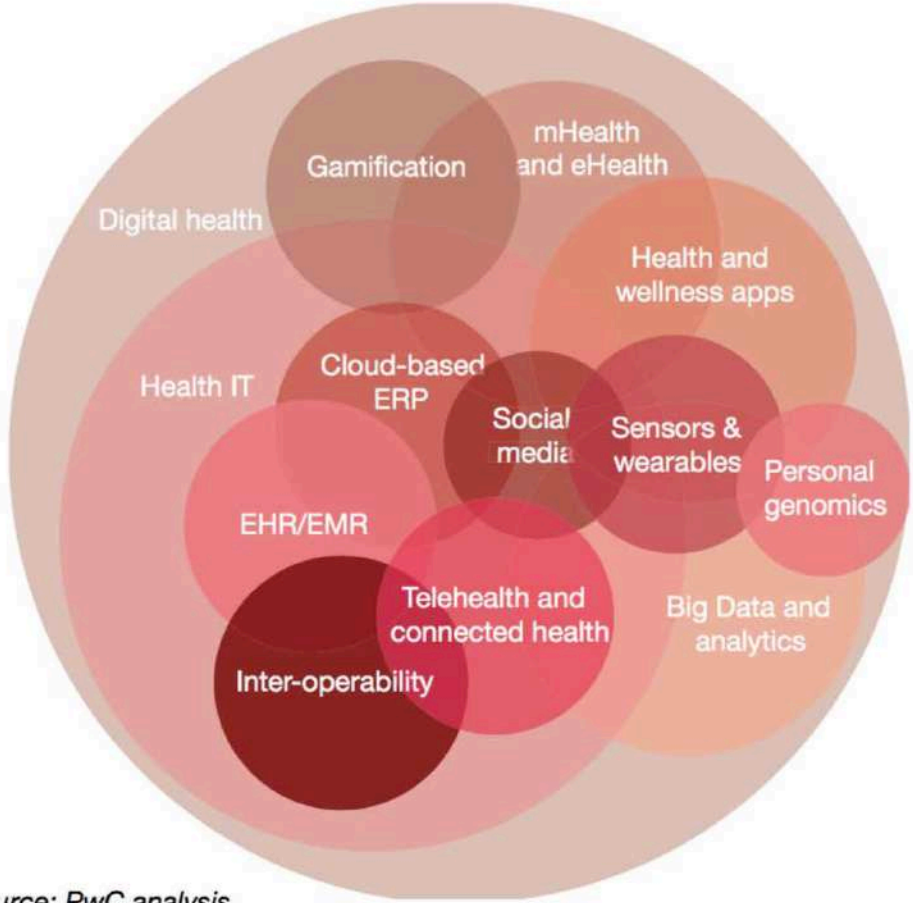
- Social media connecting healthcare
- Patient access to their own healthcare data
- Patient education

Future Healthcare Focus



Digital health ecosystem

Digital health components

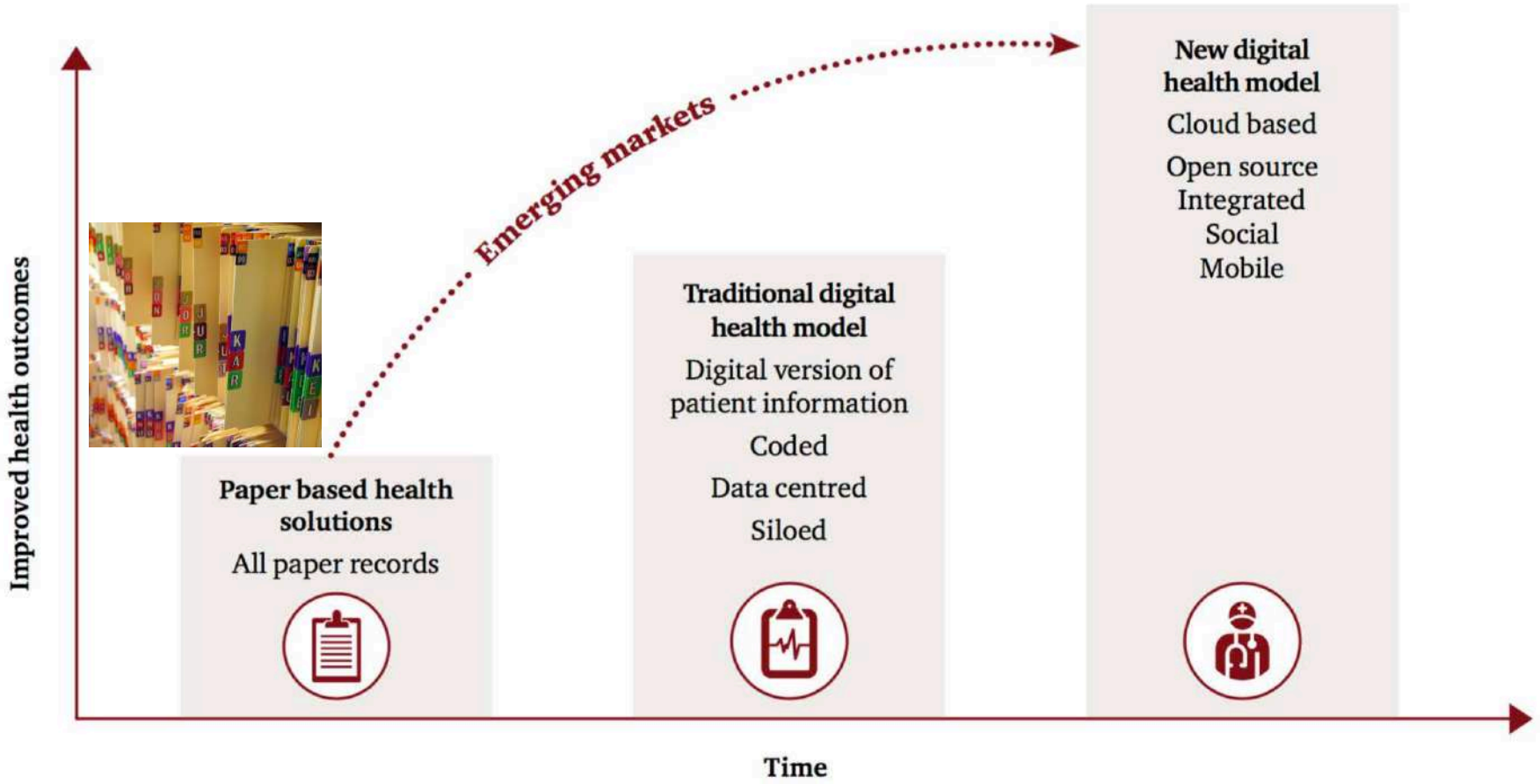


Source: PwC analysis

What does it mean for healthcare:

- Improve access
- Quality care
- Better information
- Better patient outcomes
- Increase patient engagement
- Enhance information flow

Emerging economies will leapfrog developed markets and adopt the new digital health model



New digital health model is focused on four key elements – disruption, engagement, integration and trust





Paper-based/Traditional digital health model		*New digital health model*	
Paper-based	*Digitisation*	*Digitisation*	*Disruption*
Physician-centric		*Patient-centric (Engagement)*	
Independent		*Integrated*	
Security		*Trust*	

Future Healthcare

Summary of benefits of digitising healthcare for patients, providers and payers

	Patients	Providers	Payers
EMR	Easier to read and understand	Easy storage and retrieval; improved efficiency and productivity	
EHR	Better diagnosis and treatment	Coordination and informed decision-making	Faster reimbursements
Personal Health Records	Personal wellness management	Consistency of information	Links to healthcare plans and lower claims
Remote Diagnostics	Reduces duplicated tests and referrals	Easy access	Lower cost
Remote Monitoring	Patient-centric integrated care	Reduce emergency and re-admissions	Lower cost
Telecare	Access to specialist care	Improves productivity and reduces burden of healthcare resources	Lower cost
mHealth applications	Greater patient engagement and saves time	Proactive and targeted care	
Big Data/Analytics	Accurate diagnosis, better treatment	Improves diagnostics and accuracy of treatment	Lower cost

* EMR – Electronic Medical Records; EHR – Electronic Health Records

Benefits of eHealth

More efficient work processes

Improved patient safety



Less no. of sick days & hospitalizations

Same level of access to services for all

Saved money

Bigger picture: The Digital Dream



Hospital Information System fully implemented



Patient's health record accessible at anytime



Patient's health record available centrally



Patients more involved in their treatment

This will gradually facilitate **positive lifestyle changes**

Bigger picture: The Digital Dream



Artificial Intelligence (AI) techniques will be implemented on patients' data to identify health risks



Machine Learning (ML) will enable doctors to use patients' data to make more personalized treatment decisions



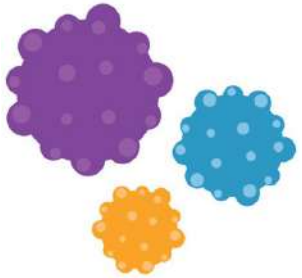
Internet of Things (IoT) is interrelated computing devices, people that make human to computer interaction

Bigger picture: The Digital Dream

Artificial Intelligence (AI) and Machine Learning (ML) will improve **cancer diagnosis** and **treatment**.



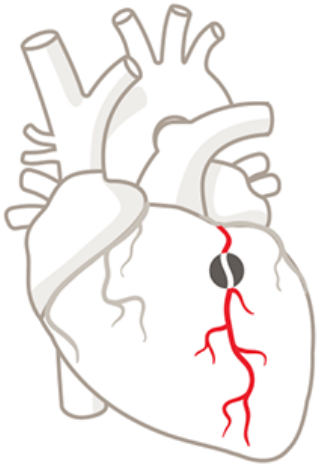
Signs of cancer detected earlier than humans

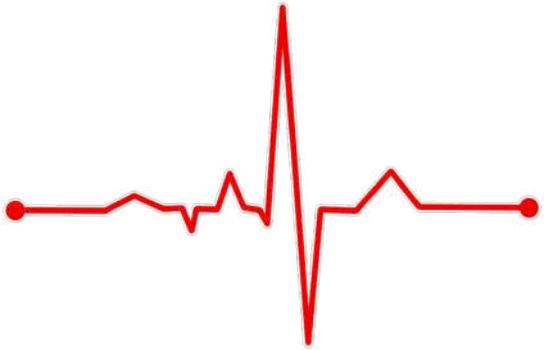


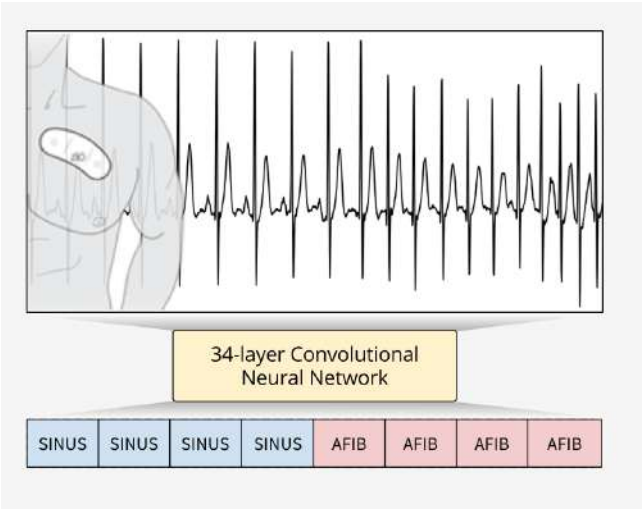
A computer system living inside human cells will **clean out afflicted cells**, upon detection

Bigger picture: The Digital Dream

Heart-attack will be predictable via ECG using **AI** and ML







Cardiologist-Level Arrhythmia Detection With Convolutional Neural Networks

Bigger picture: The Digital Dream





Brain implant can detect neural abnormalities early on or treat neuronal disorders

Bigger picture: The Digital Dream

- AI can also revolutionize **drug development**
- For e.g., Researchers* built 2 computer networks
  - 1 suggests new molecules that may have cancer-fighting properties
  - The other eliminates those suggestions based on known treatments



*Insilico Medicine, Oncotarget. 2017; 8:10883-10890.

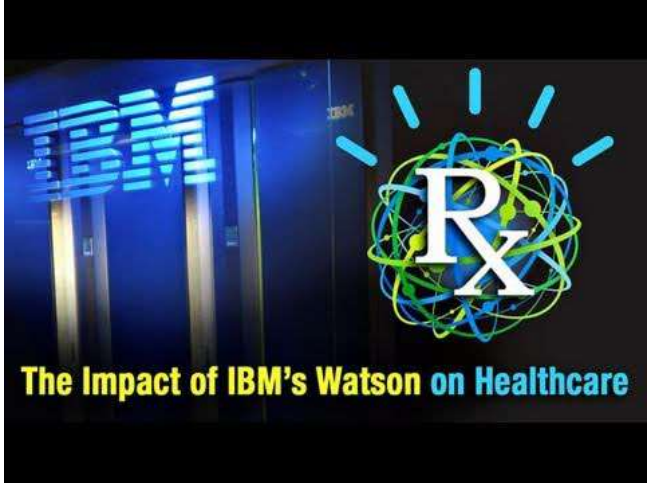
Bigger picture: The Digital Dream

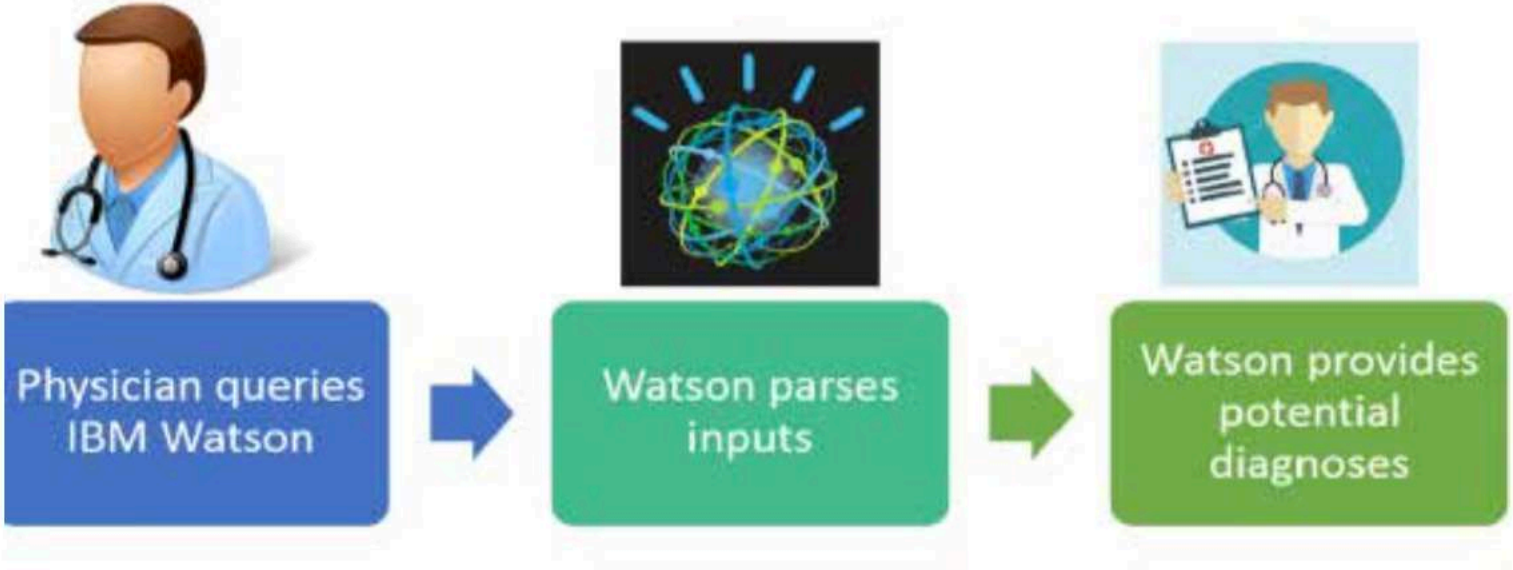




Future Healthcare

The Digital Dream





Future Healthcare

The Digital Dream :
ARTIFICIAL ORGANS



- Any machine, device, or other material that is used to replace the functions of a faulty or missing organ or other part of the human body.

- Artificial organs include the artificial heart and pacemaker (qq.v.), the use of dialysis (q.v.) to perform kidney functions, and the use of artificial substitutes for missing limbs (see prosthesis).

What are the challenges?



Adequate **funding** and support is required

What are the challenges?



New ways of working may require healthcare professionals to develop **new skills** & **knowledge**

Digital Vision: what we can do



Promote

- Preventive healthcare
- Easy to use digital methods
- Benefits of Digital Health

Future Healthcare Risks

Privacy

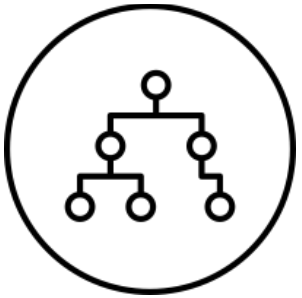


Security **Reliability**

Future Healthcare Risks

Risks?





To avoid unintended consequences, work on:

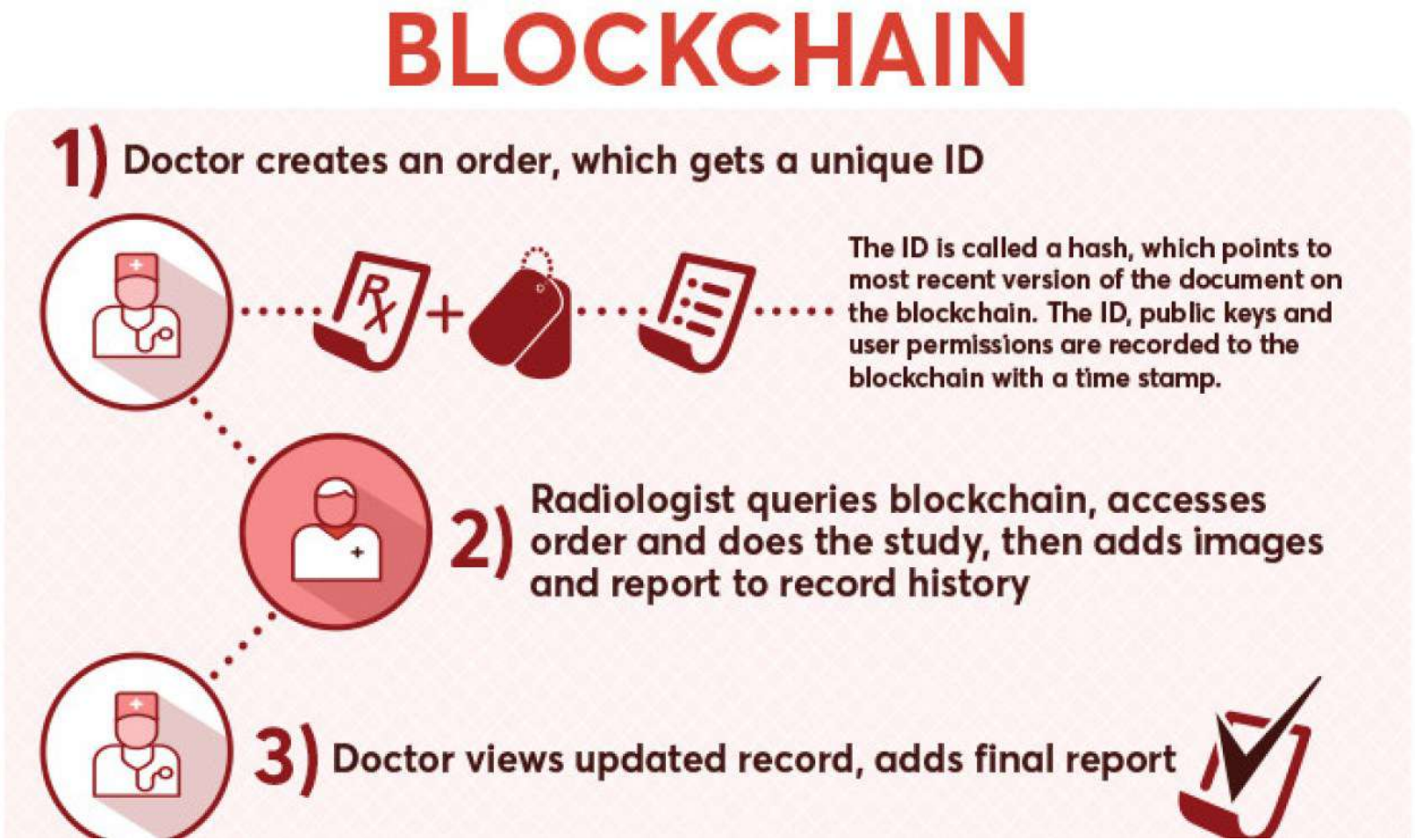
- **Strengthening methods**
- **Institutions**
- **Laws**
- **regulatory framework**

Algorithms would help via:

- **Encryption methods**
- **Privacy-preserving data processing**
- **Fraud detection systems**

Future Healthcare Risks

BLOCKCHAIN IN HEALTHCARE



Future Healthcare

Mathematical Logic

Path to Cure — Medication — Adherence — Lifestyle

$$Y = f(M, A, L)$$

$$Y = f(M, A, L)$$

Correlation= P

the **current healthcare system is broadly catering to (Y,M)** where correlation is weak

$$Y = f(M, A, L)$$

Correlation= Q

Our purpose is to **build a platform (decision system) catering to (Y, A+L)** where correlation is strong

$$P << Q$$

$$(P+Q=1)$$

Journey of AIMS Lab

AIMS Lab
Advanced Intelligent Multidisciplinary Systems Lab

UNITED INTERNATIONAL UNIVERSITY
QUEST FOR EXCELLENCE

January 2015

MediTec

INTERNATIONAL BRAIN IBRO RESEARCH ORGANIZATION

বলতে চাই

Autism Barta

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Preventive Healthcare

April 2018

More in the pipeline

AIMS Lab Research Areas

- **Research focuses on theoretical aspects and applications of the areas**

  - Intelligent Systems
  - Brain Machine Interface (BMI)
  - Human Computer Interaction (HCI)
  - Neural Engineering
  - Pattern Recognition & Machine Learning
  - Biomedical Engineering and Applications
  - ICT for Disaster Risk Reduction
  - ICT to Empower Disables
  - Rehabilitation Engineering
  - Access communication & Assistive Technologies for disables
  - Health Informatics & Mobile Technology for Health Care Applications
  - Distance Learning and Technology Enhanced Leaning

Meditec 2016





rst International Medical Engineering Conference where Doctors and Engineers are joining together to work on Healthcare issues

IBRO APRC 2017





First International Neuroscience Conference where neurologist, Neurosurgeon and Biomedical Engineers are joining together to work on neurological disorder, neuromodulation and neural interventions.

First DBS Surgery in Bangladesh

Deep Brain Stimulations (DBS)



Prof. Tipu Z. Aziz
PROFESSOR OF NEUROSURGERY,
University of Oxford, UK



After success DBS Surgery

First DBS Surgery in Bangladesh

Deep Brain Stimulations (DBS)







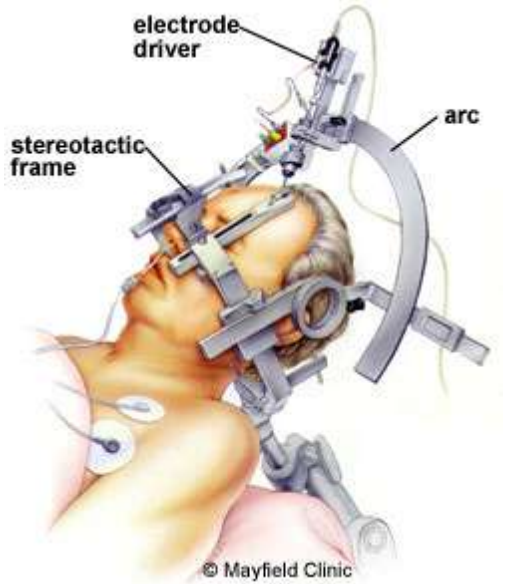


First DBS Surgery in Bangladesh

Deep Brain Stimulations (DBS)

CT Scan

MRI 1.5T



DBS Electrodes

DBS Surgeon

Two Procedures

Patient 1 : Parkinson's

Patient 2: Dystonia


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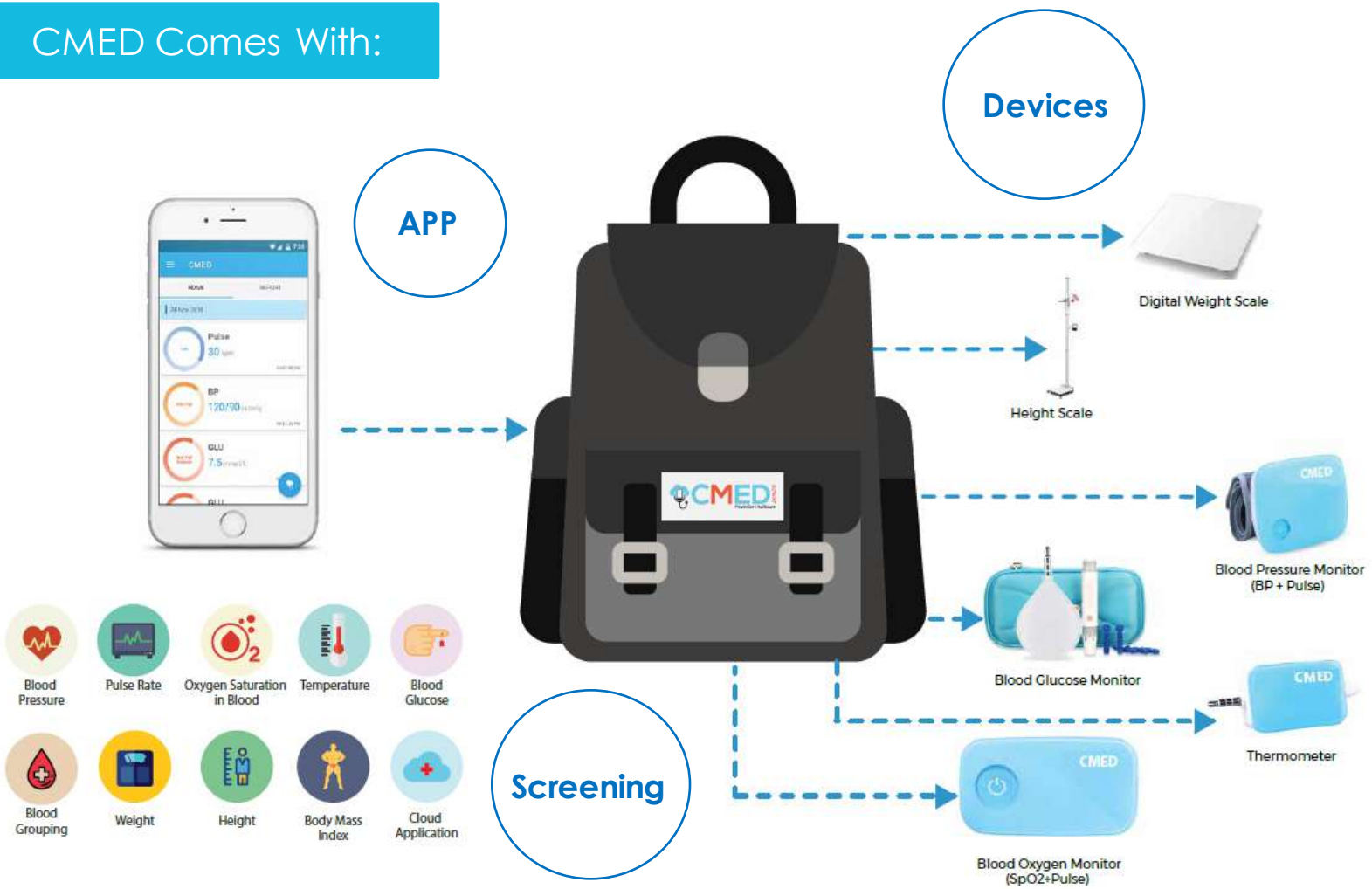
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IoT enabled AI Driven Smart Preventive Healthcare

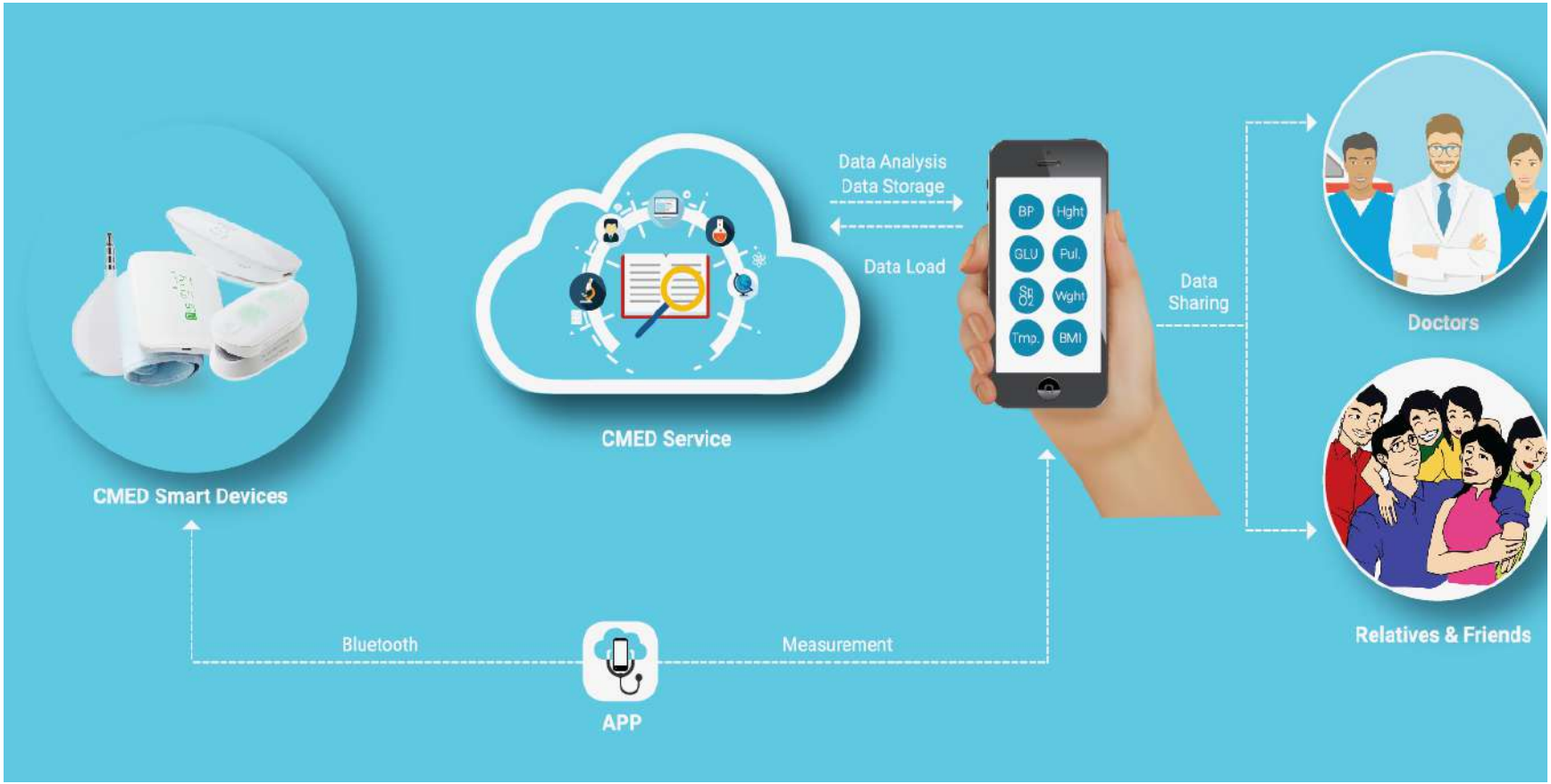
- ✓Health Monitoring
- ✓Health Risk Prediction
- ✓Ubiquitous Access to data



CMED Comes With:



How CMED System Works



Award & Recognition



Contents lists available at ScienceDirect

Computers and Electrical Engineering

journal homepage: www.elsevier.com/locate/compeleceng

CMED: Cloud based medical system framework for rural health monitoring in developing countries

Kashfia Sailunaz[a,*], Musaed Alhussein[b], Md. Shahiduzzaman[c], Farzana Anowar[d], Khondaker Abdullah Al Mamun[e]

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BASIS NATIONAL ICT AWARDS 2017

GP ACCELERATOR
Top 5

Innovation to Industry i2i

Microsoft

seedstarsdhaka*

Dhaka Winner 2017

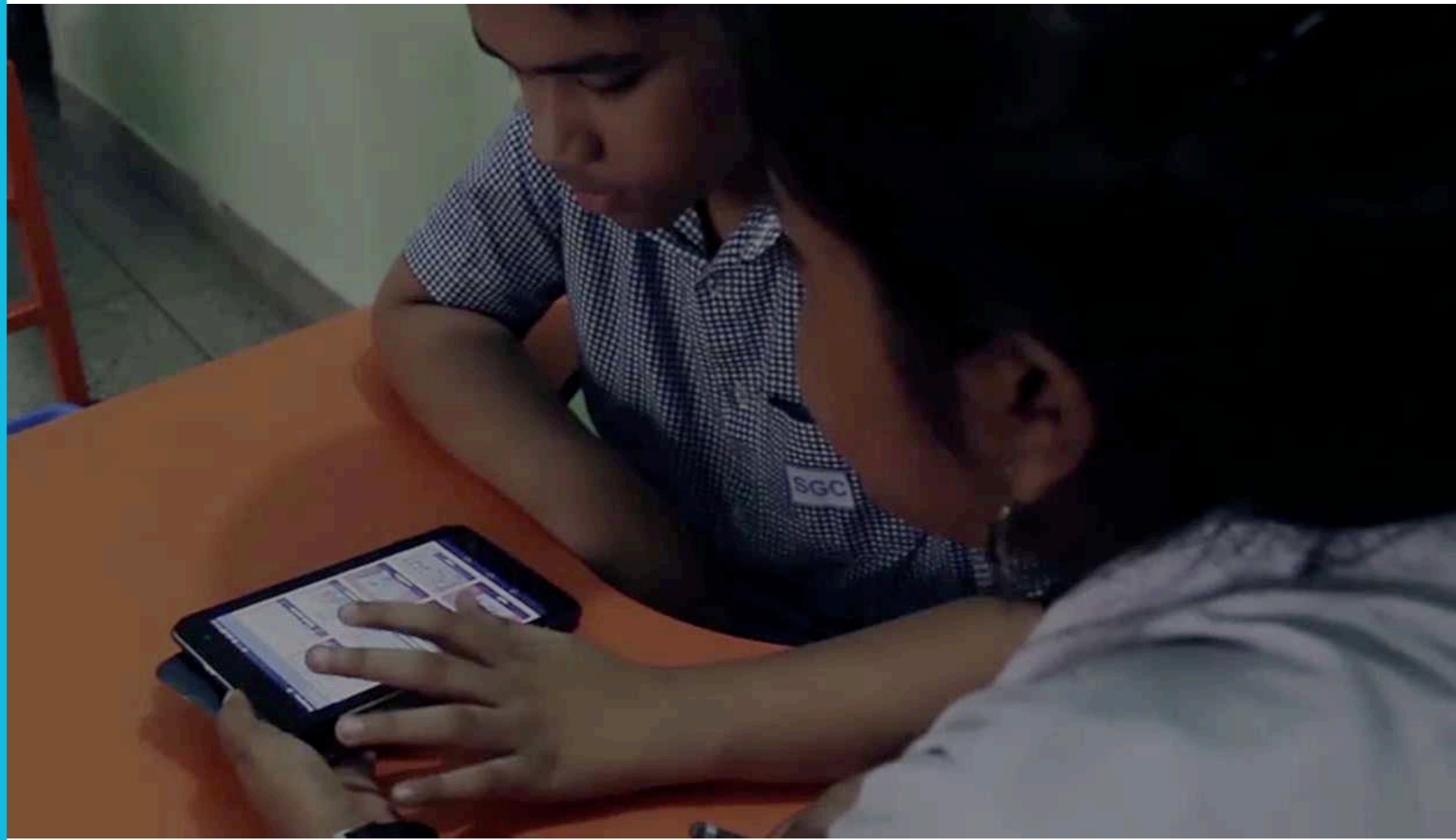
Global Innovation Winner 2018

Bolte Chai

Digital Communication
Tool for **Autistic** and
NDD Non-Verbal Children

Unique adaptability feature




SGC

জাতীয় প্রতিবন্ধী উন্নয়ন ফাউন্ডেশন
সমাজকল্যাণ মন্ত্রণালয়

AIMS Lab
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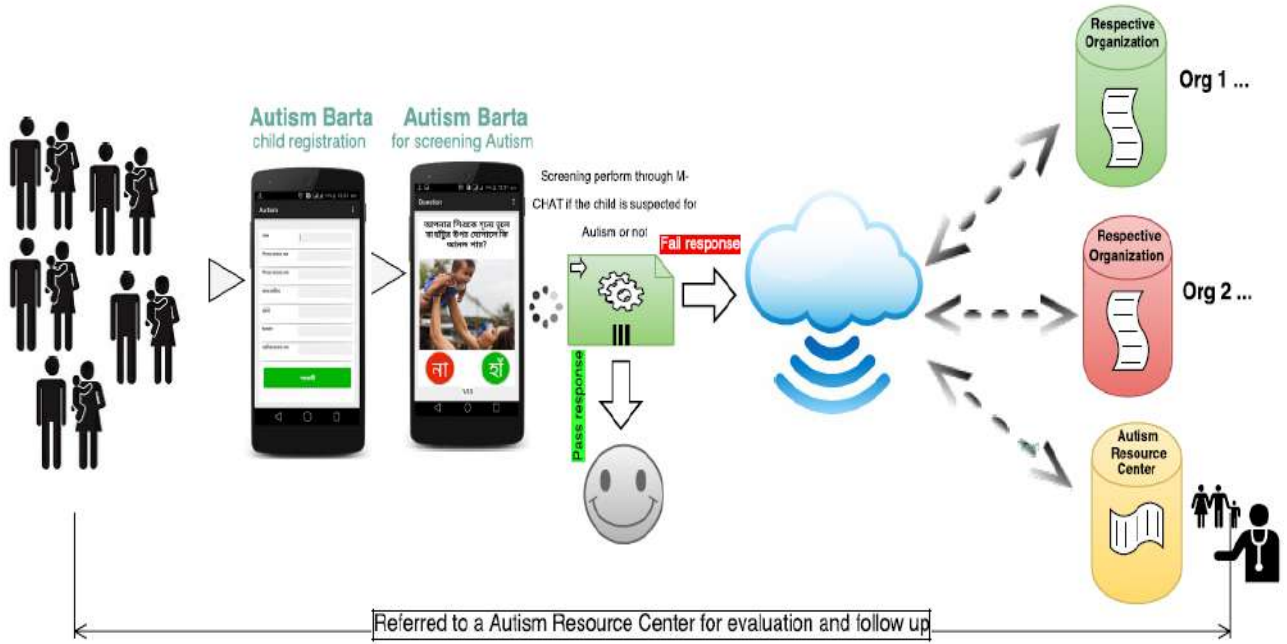
Autism Barta

Automated Autism Screening Tools

INSTITUTE FOR PAEDIATRIC NEURODISORDER & AUTISM (IPNA)
BANGABANDHU SHEIKH MUJIB MEDICAL UNIVERSITY (BSMMU)

Autism Barta: Automated Autism Screening Tools

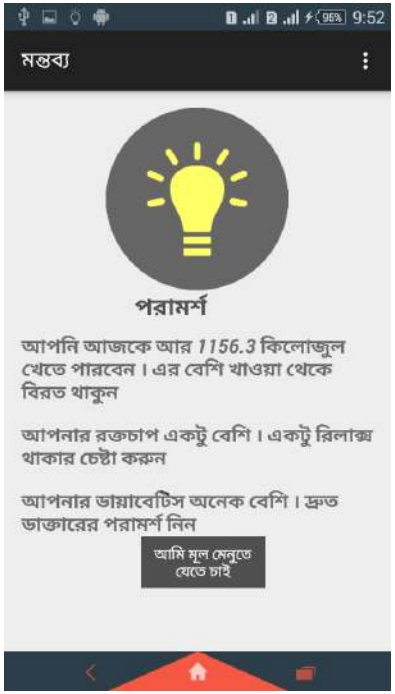




DiaHealth: AI based Diabetes Lifestyle management system











PVDoc: Parkinson's Virtual Doctor

Cloud based AI Driven Screening and Monitoring System for Parkinson's Diseases

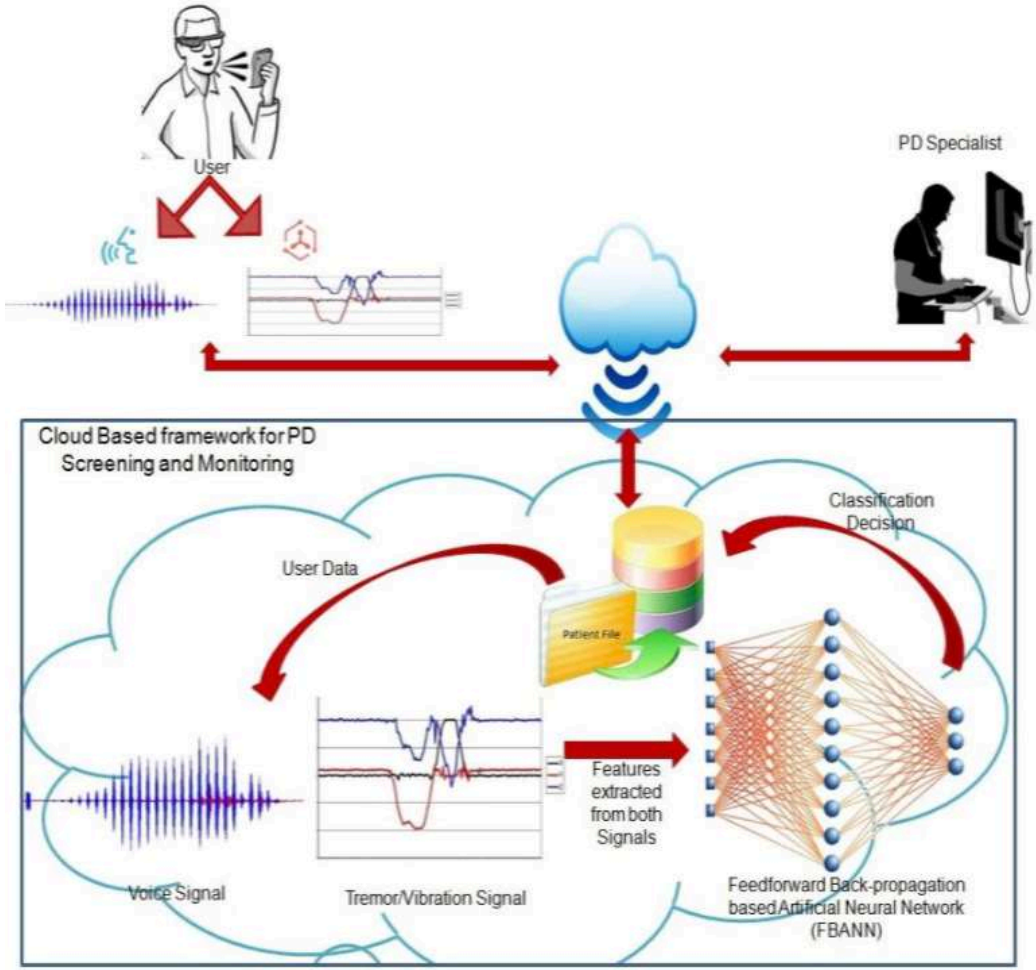
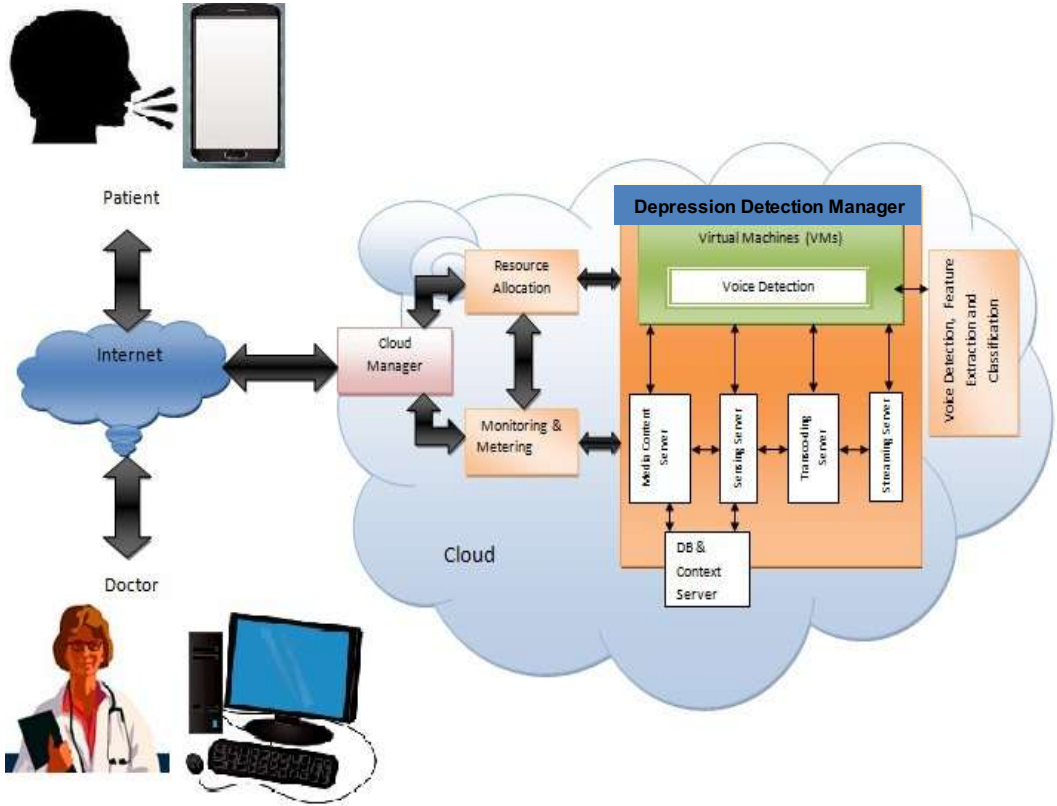


Figure 1. PVDoctor framework with its primary users.

MobiDep: Cloud based Screening, Monitoring and Rehab System for Depression







Recognitions

(National ICT Award [Health, Inclusion, R&D, Mobile App)



CMED Health



Autism Barta



Autism Barta



Bolte Chai

Recognitions

(Asia Pacific ICT Award and Global Innovation Award)

Autism Barta

Bolte Chai

CMED Health





seedstarsworld

Global Innovation Winner 2018

Dhaka Winner 2017

Acknowledgement

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Imperial College London, UK	Dr. Ravi Vaidyanathan, Dr. Michael Mace
University of Oxford, UK	Professor Tipu Aziz, Professor John Stein
Florida International University, USA	Dr. Hai Deng
King Saud University, Saudi Arabia	Dr. Ghulam Muhammad
	Professor Mohammad Nurul Huda

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UNIVERSITY OF TORONTO Holland Bloorview Kids Rehabilitation Hospital UNIVERSITY OF Southampton